

Check List Set 1

INDIVIDUAL RESEARCH COMPONENT FEASIBILITY STUDY

**GAMIFIED VR EDUCATIONAL PLATFORM FOR
MIDDLE SCHOOL CURRICULUM IN SRI LANKA**

(Technology Component)

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EXECUTIVE SUMMARY

This feasibility study evaluates the Technology component of a gamified virtual reality educational platform grounded in Sri Lankan historical technology. You focus on practical learning through interaction. You replace passive reading with task based exploration. This study documents whether the Technology module supports full development and research evaluation.

Technology education in many schools relies on textbook images and classroom explanation. Students observe systems such as irrigation outlets, storage pits, and simple machines without operating them. This limits understanding of structure, sequence, and cause and effect. Schools with limited laboratory facilities face a higher impact. Students lack repeated practice and direct feedback during learning.

Your proposed solution introduces a virtual ancient Sri Lankan village. The environment contains water tanks, controlled water outlets, food storage structures, and simple mechanical tools. You guide learners through short tasks. Each task requires observation, action, and reflection. Students adjust controls, move objects, and view system responses inside the VR space. Built in prompts and quizzes confirm learning outcomes.

The study evaluates feasibility across technical, operational, financial, educational, and ethical areas. Technical feasibility supports development using Unity and a standalone VR headset. Existing assets and interactions show readiness for full implementation. Operational feasibility supports classroom use through supervised sessions and offline deployment. Financial feasibility suits a research pilot because software tools are low cost and hardware use follows a shared model. Educational feasibility aligns with Grade 10 and Grade 11 Technology outcomes through structured tasks and assessment logging.

Findings support progression to full implementation and pilot testing. The Technology component shows readiness for controlled classroom evaluation and research data collection.

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1. INTRODUCTION

1.1 Background

This research project develops a gamified virtual reality learning platform for school level education. You work on the Technology component as an individual research contribution. The platform uses a local setting based on Sri Lankan historical practices. The focus stays on practical technology learning through interaction.

Technology subjects require learners to understand systems, tools, and processes. Many schools teach these topics through drawings and verbal explanation. Students observe diagrams of irrigation outlets, storage pits, and simple machines. They do not operate these systems. This limits understanding of how parts connect and how actions change outcomes.

The current VR environment includes a village area, agricultural land, animals, and a basic water system. Your Technology component extends this environment into structured learning activities. Each activity links a visible system to a learning outcome. You design tasks where learners interact with objects, follow steps, and view results. This approach supports skill development through action.

The Technology component supports Grade 10 and Grade 11 learning needs. It introduces water management, storage methods, and basic mechanical tools through guided tasks. You design the module for classroom use with limited hardware. The design supports short sessions and repeated use.

1.2 Objective of the feasibility study

This feasibility study checks whether the Technology component supports full development and classroom use. You review readiness before implementation and evaluation. The study focuses on the following areas.

- Technical feasibility of VR interactions, performance, and stability
- Operational feasibility for school level deployment and supervision
- Financial feasibility for a research pilot setting
- Educational feasibility for curriculum alignment and assessment
- Ethical and environmental feasibility for safe and responsible use

The study guides decisions on implementation scope, risk control, and evaluation planning.

1.3 Scope of the study

This document evaluates only the Technology component of the VR platform. It does not cover Science, Mathematics, or Engineering components. The scope includes the design of learning tasks, interaction methods, deployment approach, and data collection for research evaluation.

The study assumes use of a standalone VR headset in a supervised classroom setting. It focuses on feasibility for a pilot study rather than large scale deployment. The scope includes constraints related to hardware access, session time, and user comfort.

2. PROBLEM STATEMENT

Technology education aims to develop understanding of systems, tools, and processes through observation and practice. In many school settings, this goal remains unmet. Lessons rely on textbook diagrams, board drawings, and short explanations. Students rarely interact with working models. This gap affects understanding and skill development.

Topics such as irrigation systems, food storage methods, and simple machines require learners to see movement, pressure, balance, and force. When students only view static images, they learn names and labels. They do not understand operation steps or cause and effect. This leads to surface level learning and weak problem solving ability.

Limited laboratory access increases the problem. Many schools lack physical models or tools for repeated practice. Teachers face time limits and safety concerns. As a result, practical activities become demonstrations rather than hands on tasks. Students observe instead of acting.

The absence of local context adds another issue. Many examples in books do not reflect Sri Lankan environments or historical practices. Students fail to relate lessons to real situations. This reduces engagement and retention.

A practical learning solution must meet several needs.

- Support repeated practice without material loss
- Show system behavior through direct interaction
- Fit classrooms with limited resources
- Reflect local technology and daily life

The current teaching approach does not meet these needs. This study examines whether a VR based Technology module provides a feasible solution.

3. PROJECT DESCRIPTION

3.1 Overview of the proposed solution

The Technology component forms a structured learning module inside a virtual ancient Sri Lankan village. You design this module to teach technology concepts through direct interaction. The environment represents daily life systems used in farming and village activities. Each system supports a specific learning outcome.

The module follows task based learning. You guide the learner through short activities. Each activity focuses on one technology concept. Learners observe the system first. They then perform actions. The system responds with visible changes. This process supports understanding through action.

The module design supports short VR sessions. Each task runs for a limited time. This approach fits classroom rotation and reduces user discomfort. Teachers guide the session and monitor progress.

3.2 Learning context and alignment

The Technology component aligns with Grade 10 and Grade 11 Technology subject areas. The focus stays on practical understanding rather than theory memorization. Each VR task links to a real world system used in Sri Lankan history and daily life.

The module supports the following learning areas.

- Water management and controlled water release
- Storage methods for food preservation
- Basic mechanical systems and simple machines

You design tasks to support observation, operation, and reflection. Learners answer short questions after completing each task. This confirms understanding and supports assessment.

3.4 Key functional features

The Technology module includes several core features. Each feature supports interaction and feedback.

- Water tank interaction to store and release water
- Flow control activity to observe controlled water movement
- Storage system inspection to identify design features

- Simple machine tasks using pulleys and levers
- Guided prompts to explain steps and safety
- In scene quizzes to confirm learning outcomes

Each feature uses simple interaction methods. You use grab, pull, rotate, and press actions. Visual cues guide the learner during tasks.

3.5 Technology learning activities

Water management activity: The learner controls a water outlet connected to a tank. The learner adjusts the outlet and observes water flow. The system shows how controlled release affects irrigation.

Storage systems activity: The learner inspects a storage structure such as a granary or pit. The learner identifies features linked to protection from moisture and pests. The task ends with short questions.

Simple machines activity: The learner uses a pulley or lever to move an object. The system shows effort and movement direction. The learner completes the task within a time limit.

3.6 User interaction design

Interaction design focuses on clarity and safety. The learner uses teleport movement. This reduces discomfort. Each task begins with a short instruction panel. Prompts appear when action is required. Feedback appears after each step.

The system avoids complex controls. Each task uses a small number of actions. This supports ease of use for first time users.

4. FEASIBILITY ANALYSIS

4.1 Technical feasibility

Platform and tools

The Technology component uses a VR ready development stack suitable for school level deployment. You select tools based on stability, support, and ease of maintenance. The platform supports standalone operation and offline sessions.

- Unity engine for scene control and interaction logic
- XR interaction framework for grab, pull, rotate, and press actions
- Standalone VR headset for untethered movement
- Optimized 3D assets for village structures and tools
- Local data storage for session logs and quiz results

These tools support rapid development and controlled testing. You already use the environment in earlier stages of the project. This reduces setup risk.

Interaction implementation

Each learning task uses simple interactions. You avoid complex gestures. The system relies on clear input rules.

- Grab and release for tools and handles
- Pull and rotate for outlets, levers, and gates
- Press actions for confirmations and quizzes
- Visual highlights to guide required actions

This approach supports first time users. It also reduces error rates during tasks.

Performance and comfort

Standalone VR hardware has limits on processing power. You manage performance through design choices.

- Low polygon models for large structures
- Baked lighting to reduce real time load
- Limited particle use for water visuals
- Teleport movement to reduce discomfort
- Short session design to manage fatigue

These controls support stable frame rates and user comfort.

Technical risks and controls

The main technical risks relate to interaction accuracy and performance stability.

- Interaction failure risk reduced through snap points
- Frame drop risk reduced through profiling and asset optimization
- Data loss risk reduced through local save after each task

4.2 Market feasibility

Target users

The primary users are Grade 10 and Grade 11 students who study Technology subjects. The secondary users are Technology teachers who guide sessions and review learning outcomes.

User need analysis

Schools face several constraints that support demand for this solution.

- Limited access to physical technology models
- Limited lab time for practical activities
- High student interest in interactive learning tools
- Need for curriculum aligned practical activities

The Technology component addresses these needs through reusable virtual tasks.

Existing alternatives

Existing alternatives include textbook diagrams, videos, and slide presentations. These tools show structure but not operation. They do not support hands on action. They do not provide local historical context. The VR module fills this gap through direct interaction and observation.

4.3 Financial feasibility

Cost structure

The project targets a research pilot rather than commercial deployment. Costs remain limited and controlled.

- Standalone VR headset for testing and sessions
- Development workstation and standard peripherals
- Optional asset purchase for specific models

- Transport and printing for evaluation activities

Most software tools use student or academic licenses. This keeps costs low.

Value justification

The module supports repeated use without material wear. One headset supports many learners through rotation. The same module supports multiple lessons and evaluation sessions. Financial feasibility supports continuation.

4.4 Operational feasibility

Deployment model

The module follows a teacher guided classroom model. One headset supports rotation sessions.

- Teacher explains task goals and safety rules
- Student completes a short VR task
- System saves task results locally

This model fits typical classroom constraints.

Session management

Each VR task runs for a limited time. You design tasks to complete within 10 to 15 minutes. This supports multiple students per session and reduces fatigue.

Scalability

Scalability depends on hardware access. Schools with more headsets run parallel sessions. Schools with one headset use rotation. Tasks operate independently. This supports flexible deployment.

4.5 Environmental and ethical feasibility

The Technology component promotes responsible resource use through water management and storage topics. The content respects cultural context and historical practices.

- Supervised VR use to ensure safety
- Clear play boundaries to avoid physical risk
- Anonymous user IDs for data logging
- Consent based participation for evaluation

The module avoids harmful content and supports safe learning conditions.

5. RISK ANALYSIS

5.1 Risk identification

This project involves technical systems, classroom use, and student interaction. Each area carries specific risks. You identify risks early to reduce impact during implementation and evaluation. The risks fall into technical, operational, educational, and data handling categories.

5.2 Technical risks

Technical risks relate to VR performance, interaction reliability, and hardware limits.

- VR discomfort due to motion or session length
- Frame rate drops on standalone hardware
- Interaction errors during grab or pull actions
- Application crash or unexpected behavior

You reduce these risks through controlled design. Teleport movement reduces discomfort. Short tasks limit fatigue. Optimized assets support stable frame rates. Snap points and visual cues improve interaction accuracy. Local saves reduce data loss during crashes.

5.3 Operational risks

Operational risks affect classroom deployment and supervision.

- Limited time per student during rotation
- Hardware damage during handling
- Classroom space constraints
- Teacher unfamiliarity with VR setup

You manage these risks through clear session rules. Teachers supervise all sessions. Students receive brief handling instructions before use. The system uses a simple start and exits flow. Tasks run within fixed time limits.

5.4 Educational risks

Educational risks affect learning outcomes and research validity.

- Learners focus on interaction rather than learning goals

- Tasks fail to match curriculum outcomes
- Assessment questions lack clarity

You control these risks through task design. Each activity links to a clear learning outcome. Prompts guide learners toward correct actions. Short quizzes confirm understanding before task completion.

5.6 Risk summary table

Risk	Likelihood	Impact	Mitigation
VR discomfort	Medium	Medium	Teleport movement and short sessions
Low frame rate	Medium	High	Asset optimization and profiling
Interaction failure	Medium	Medium	Snap points and visual guidance
Hardware damage	Medium	Medium	Supervised use and handling rules
Low learning gain	Medium	High	Clear outcomes and guided tasks

6. RECOMMENDATIONS

6.1 Summary of feasibility findings

The feasibility review supports progression to full development and pilot evaluation of the Technology component. Technical review confirms platform readiness using Unity and a standalone VR headset. Interaction design supports stable performance and user comfort. Operational review supports classroom use through short sessions and supervised rotation. Financial review supports a pilot model with low software cost and shared hardware. Educational review supports alignment with Grade 10 and Grade 11 Technology outcomes through task based learning and in scene assessment.

These findings indicate readiness for controlled implementation and evaluation.

6.2 Design and development recommendations

Focus development on clarity, stability, and learning outcomes.

- Finalize a clear task list with one outcome per task
- Build a short tutorial zone for first time users
- Use simple interactions with clear visual guidance
- Keep task duration within 10 to 15 minutes
- Add immediate feedback after each action

These steps reduce learning friction and improve consistency across sessions.

6.3 Deployment recommendations

Adopt a teacher guided deployment model for pilot testing.

- Use one headset with rotation for initial trials
- Schedule sessions within regular lesson periods
- Provide a short teacher guide for setup and control
- Enforce clear safety and handling rules

This approach fits typical school constraints and supports supervision.

6.4 Evaluation and research recommendations

Plan evaluation activities early to support data quality.

- Define clear metrics for task completion and accuracy
- Use short quizzes linked to each task
- Log time on task and action errors
- Collect student feedback through simple questionnaires

These steps support valid data collection and analysis.

6.5 Future enhancement recommendations

After pilot testing, consider controlled expansion.

- Add more depth to water management scenarios
- Introduce comparative storage designs
- Expand simple machine challenges with variation
- Improve visual explanations through overlays

7. CONCLUSION

This feasibility study evaluated the Technology component of a gamified virtual reality educational platform grounded in Sri Lankan historical practices. The study reviewed technical, operational, financial, educational, environmental, and ethical aspects. Each area supports progression to full implementation under controlled conditions.

The Technology component addresses a clear gap in practical learning. It allows learners to interact with water systems, storage structures, and simple machines. The design supports observation, action, and feedback. Short tasks and guided prompts support classroom use and user comfort.

Technical readiness appears strong due to stable tools and optimized interaction design. Operational readiness supports supervised classroom sessions with limited hardware. Financial readiness supports a research pilot through low software cost and shared equipment. Educational readiness supports curriculum alignment and measurable outcomes.

The next phase focuses on implementation completion and pilot evaluation. Results from controlled classroom testing guide refinement and future expansion. The Technology component shows strong potential as a practical learning tool and a valid research contribution.

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